Influence of environmental factors and phytoplankton blooms on cyphonautes larvae abundance and bryozoan colony development in kelp farms of mid-Norway.

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1. Abstract (250 words)

Seaweed cultivation offers a sustainable solution for biomass production. However, in Norway, biomass quality at kelp farms is affected by biofouling, typically from encrusting bryozoans, such as *Membranipora membranacea* and *Electra pilosa.* This study investigated the drivers of bryozoan biofouling at a kelp farm in the coast of mid-Norway in 2022 and 2023. Environmental parameters (temperature, salinity, turbidity, light, nutrients and wind), phytoplankton concentrations (chlorophyll *a* and size structure), and bryozoan (cyphonautes) larval size, abundance and recruitment on the kelp species, *Saccharina latissima*, were monitored. Spring phytoplankton blooms (up to ~ 6 mg chlorophyll a m^{-3}) followed increased irradiance and reduced mixing, with larvae showing two main abundance peaks – in April (~ 200 - 400 ind mm^{-3}), 1-2 weeks after the onset of the bloom - and in June (~ 450 ind mm^{-3}). Larval abundance was associated with low salinity (value ~ 32), stratified, fresher coastal waters. *Membranipora membranacea* larvae were generally more abundant and reached larger sizes (up to 0.6 mm in length) during the spring settlement period (late April-June). Colony abundance, size, and areal coverage were higher in earlier-deployed kelp (October 2022 versus January 2023) and increased exponentially from May (<1%) to late June (up to 11%). Recruitment peaked during a sharp increase in temperature (1-2°C in a few days) and was low during a period with high wind speed (up to 15 m s^{-1}). Our findings demonstrate that food availability, water column stability, and rapid temperature increases are the dominant factors influencing bryozoan biofouling on kelp.

2. Introduction

There is a need for sustainably sourced human food and animal feed (FAO, 2022). Cultivated seaweed can be a sustainable biomass which contributes to food security, due to its high nutritional value and low environmental impact and CO_2 footprint (Imchen, 2021; Koesling et al., 2021; Thomas et al., 2021). While the industry is well-established in Asia, seaweed sea cultivation at sea has been gaining momentum in Europe over the last 10 years, particularly for the brown macroalgal kelp species *Saccharina latissima* ((Linnaeus) C.E.Lane, C.Mayes, Druehl & G.W.Saunders 2006) and *Alaria esculenta* ((Linnaeus) Graville 1830) (Afonso et al., 2021; Forbord et al., 2020). However, the industry is still very much in its infancy compared to Asia, and seaweed cultivation in Europe and Norway is still not profitable at the current scale largely because of biofouling (Moscicki et al., 2025).

Biofouling by the encrusting colonies of bryozoan species *Membranipora membranacea* (Linnaeus, 1767) and *Electra pilosa* (Linnaeus, 1761) is one of the main factors that limits high quality yield of sea cultivated kelp (Skjermo et al., 2014). The rapid appearance of bryozoan colonies on the kelp fronds (days to weeks in spring/summer) damages the kelp lamina and reduces its commercial value, forcing the seaweed farmers to cut short their harvest when kelp growth is at its peak (Førde et al., 2016). In Norway, settlement and colonial growth of *M. membranacea* and *E. pilosa* on cultivated kelp vary according to latitudinal gradient, following the rise of temperature and availability of food (phytoplankton blooms) (Forbord et al., 2020; Førde et al., 2016). However, biofouling coverage can increase rapidly, and the onset can vary from year to year, even within farms from the same region. This makes it difficult for planning of harvest timing, and optimising biomass yield before quality is compromised (Njåstad, 2018).

The life cycle of *M. membranacea* and *E. pilosa* consists of two distinct life stages, one as microscopic planktotrophic larvae, known as cyphonautes, and another as an encrusting colony that grows on ephemeral substrates, such as the kelp lamina (Woollacott and Zimmer, 2013). After sexual reproduction, larval populations of *M. membranacea* and *E. pilosa* cyphonautes feed on phytoplankton and are, therefore, able to remain in the water for days or weeks before settling on the kelp (Woollacott and Zimmer, 2013). After finding a suitable substrate, the larva settles and undergoes metamorphosis into a founding zooid (ancestrula), which gives rise to the colony through rapid asexual budding (Woollacott and Zimmer, 2013). A yearly monitoring program that quantifies cyphonautes abundance and maturation state in the waters, and also relates their presence with other environmental factors, such as temperature changes and the timing of phytoplankton blooms, could provide an early warning for predicting when biofouling will start. Likewise, frequent monitoring of the early stages of bryozoan colonies in kelp, which are often not captured with the naked eye, is crucial for predicting the initial growth and expansion of the colonies in a kelp farm (Cohen, 2019; Førde et al., 2016; Matsson et al., 2019).

Temperature and water column stratification have been suggested as the main factors that induce larval recruitment and control the growth of bryozoan colonies (Saunders and Metaxas, 2010). Light triggers photosynthesis and growth of phytoplankton, the main food source of bryozoan larvae, and has been suggested also to induce larval hatching (Woollacott and Zimmer, 2013). Other environmental factors and cues, such as food availability (using phytoplankton abundance as a proxy) (Woollacott and Zimmer, 2013), chemical signalling from bacterial biofilms and spectral changes in light might also play a role. Chemical cues released by macroalgae have been suggested to induce the transition from cyphonaute larvae of *M. membranacea* in the water column and settlement to the benthic habitat (Seed and O'Connor, 1981). The timing and concentration of the spring phytoplankton bloom likely influences cyphonautes abundance, settlement and colony growth. Likewise, phytoplankton species composition, specifically dominated by small flagellates, has been shown to increase bryozoan coverage on kelp and other substrates (Bullivant, 1968; Njåstad, 2018). Yet, it is unclear how these factors interact and trigger the sudden settlement and growth of bryozoans on lamina (days to a few weeks). A frequent, accurate monitoring of the early stages of bryozoan colony development, in addition to environmental parameters would elucidate the factors involved in the growth and survival of biofouling organisms.

This paper aims to understand the pillars of bryozoan colonies appearance and growth on a kelp farm from mid-Norway by assessing, throughout the cultivation period, the following parameters: 1) changes in environmental factors (temperature, salinity, irradiance, dissolved inorganic nutrients, turbidity and wind speed), 2) changes in the timing and size composition of the phytoplankton spring bloom, 3) abundance and size of cyphonautes in the water and 4) timing of larval settlement, recruitment and growth on a kelp lamina. The environmental factors, phytoplankton cell size and larval abundance were monitored within two cultivation years (2022 and 2023). Additionally, within one cultivation period (2023), we used a low-cost flow through microscope and flatbed photo scanners to digitally visualize bryozoans and to provide detailed information related to larval size in the water column and colony size and growth on the kelp. The questions we aimed to answer with this study were: Does the timing and size composition of the phytoplankton blooms influence the start of biofouling? Can larval abundance and size serve as a proxy for early biofouling detection? The results of this research provide insights into which environmental factors to consider when predicting the appearance and growth rate of bryozoans on the kelp lamina. The results might also help farmers plan their operations and avoid unnecessary loss of biomass or economic value due to biofouling.

3. Material and methods

3.1 Study site

The present study was conducted at a kelp farm at Frøya island, off the coast of Central Norway, Trøndelag (Fig. 1), from February to June 2022 and 2023. The farm (Måsskjæra) belongs to Seaweed Solutions (SES) (63°44.617'N 8°52.756'E). The fieldwork was done at two sites, one by the edge of the seaweed farm (station MI) and another outside (station MO), approximately 500-700 m from MI (Fig. 1c). While the location of station MO was the same during both years, the location of station MI at the edge of the cultivation frame moved ~100 m from 2022 to 2023 due to farm logistics (Fig. 1c). The reason we chose these sites is that we wanted to collect larval samples from stations at the kelp cultivation site as well as a ~ 100 m away, to evaluate the spatio-temporal variability of the larvae in the kelp farm surroundings.

The Frohavet area is known to be biologically productive with intensive fishing and aquaculture activities (Ervik et al., 2018). The oceanography in the area is characterized by complex currents due to strong winds, shallow and uneven bathymetry, and water mixing events (from tides and storms) (Sætre, 2007). Two major currents are found along the coast of Norway - the North Atlantic Current (NAC) – a warm and nutrient-rich water mass that lies beneath the Norwegian Coastal Current (NCC) - a relatively low salinity water mass (Sætre, 2007). The NCC is comprised by freshwater runoff from fjords (including the Trondheimsfjord) that accumulates along the coast of Norway, particularly from spring to autumn. The seabed topography is comprised of mixed substrates, with large rock structures separated by areas with soft sediments (Keeley et al., 2019).

Måsskjæra is a semi-exposed kelp farm with depth ranging from 8 to 35 m. The location is exposed to north-easterly winds and sheltered from westerly and southerly winds (Førde et al., 2016). The cultivation site is approximately 440 × 440 m and is based on a horizontal longline system divided into approximately 16 cultivation squares (Overrein et al., 2024) (Figure 1c). *Saccharina latissima* (sugar kelp) is one of the main cultivated species at the SES farm and this species was seeded onto a 2 mm twine (polyester) at SES hatchery in Trondheim. Seedlings were deployed with a size of 1 cm. The seeded lines were spun around a 20 m stretch of carrying rope of 14 mm diameter. The distances between each line were 1.5 m. The cultivation depth was around 3 m depth.

3.2 Environmental data collection using sensors

Environmental data were collected to investigate if, and to what extent, they affect cyphonautes abundances. To record the physical properties of the water column at each station, vertical profiles (down to 10 - 15 m) of conductivity (later converted to salinity), temperature (°C), turbidity ($Turb_{CTD}$, Formazin Turbidity Units (FTU)) and chlorophyll *a* fluorescence ($Chl\textit{a}_{CTD}$, mg m^{-3}) were performed at each station (MI and MO) using a CTD device (model SD204 SAIV A/S™).

In addition to vertical profiles with a CTD, a C3 submersible fluorometer sensor (Turner Designs, USA) was deployed at 3 m depth and attached to the mooring frame of the farm at station MI-22 in 2022 and MI-23 in 2023 (Fig 1c). The sensor collected time-series data (every 10 minutes) of chlorophyll a fluorescence ($Chl\textit{a}_{frame}$, later converted to biomass values, mg m^{-3}), turbidity ($Turb_{frame}$, FTU) and temperature (°C) from February to June in 2022 and 2023. The sensor had an antifouling copper plate and a mechanical wiper that rotated and cleaned the optical sensors before each measurement was taken.

To visualize the overall trend of above-water, solar radiation available throughout the cultivation season, satellite-derived all-sky surface total Photosynthetically Active Radiation (PAR_{SAT}, 400-700 nm) was obtained from the NASA Power's online database from Måsskjæra's site (NASA POWER, 2024). Wind speed data (m s^{-1}) was extracted from the Sula meteorological station in western Frohavet (Fig. 1a) and from the Norwegian Centre for Climate Services (https://seklima.met.no/).

3.3 Water and net sampling

Water and net sampling (more details below) were taken at both sites over the two years. Sampling was done once or twice a month from February to June 2023 and 2024, adding up to 17 days in total (8 days in 2022 and 9 days in 2023). Selection of sampling days was partially dependent on weather and tidal conditions (preferably low tides and weak wind conditions for consistency), as well as personnel availability.

Seawater samples were collected with a custom-made 5 L water sampler, at 3 m depth at each site. The water was stored in two 8 L acid-washed dark bottles and filtered at the dock a few hours later for nutrients, and phytoplankton size structure.

For nutrient concentrations (nitrate plus nitrite [NO_2^- + NO_3^-] - hereafter referred as to nitrate [NO_3^-],) as well as ammonium [NH_4^+], phosphate [PO_4^{3-}] and silicate [$Si(OH)_4$]), triplicate water samples from each station were filtered with a 0.8 µm polycarbonate filter (Whatmann), after being flushed to remove artificial ammonium residuals. The filtrate (~ 45 ml of each sample) was placed into centrifuge tubes and kept on dry ice for transportation and stored at -20 °C in the freezer until further analysis.

For phytoplankton size structure, water samples (~10 mL) were fixed with pre-filtered 25% glutaraldehyde (Fisher Scientific U.K.) at a final concentration of 0.25%. The samples were then wrapped in aluminium foil and after 15–30 min of fixation stored on dry ice until transportation to a -80°C freezer at Trondheim Biological Station (TBS), according to Fragoso et al. (2019b).

Zooplankton net sampling was conducted for estimating cyphonautes abundance. Vertical hauls of zooplankton from the surface (upper 5 m) were taken using a handheld zooplankton net (100 µm mesh size and 40 cm in diameter, Hydrobios) and a filtering cod-end. Triplicate samples were collected at most stations and sampling dates. The net was rinsed with 96% ethanol (EtOH) each time and preserved in this solvent in 200 mL plastic bottles and stored dark and in room temperature for later analysis.

3.4 Kelp sampling for bryozoan colony identification

The quantification of bryozoan coverage on the kelp lamina from digital scans was done only from *S. latissima* specimens cultivated in 2023 from station MI-23, while in 2022, only visual inspections (but no digitalization) were done. The reason for this is that a protocol for bryozoan detection was applied in 2022, but it was unsuccessful for identification of small colonies (Thomason, 2023). In 2023 a new methodology was developed, where cyphonautes larvae, ancestrula and small to large colonies in the kelp lamina were detected using a flatbed scanner (Snijder, 2024). For this, *S. latissima* specimens were collected from two distinct cultivation lines (seedlings deployed in October 2022 and January 2023) every 1-2 weeks at seven sampling dates subject to weather and boat availability from late April until the end of June 2023. On every sampling date, five specimens from each deployment line were randomly selected (Table S1, supplementary material). The kelp samples were rolled up in aluminium foil, put into plastic bags, stored in dry ice (-80 °C) for transportation, and later stored in a freezer (-20°C) at TBS, Trondheim, Norway.

3.5 Laboratory analysis of nutrient concentrations

Filtered seawater samples for [NO_3^-], [$Si(OH)_4$], [PO_3^{4-}] and [NH_4^+] were analyzed at TBS. Analysis for dissolved nutrients in water samples were done by using Flow Solution IV system O.I. Analytical Auto analyzer according to the Norwegian standard 4724 and 4745 respectively. Determination of NH_4 was done by direct segment flow analysis according to Kérouel and Aminot (1997).

3.6 Phytoplankton size structure

Samples were analyzed within 24 months of collection using a CytoSense benchtop flow cytometer (CytoBuoy, b.v., NL) equipped with a blue laser (488 nm, 50 mW) to allow discrimination between phototrophic and non-phototrophic particles. The sheath fluid consisted of Milli-Q water and filtered through a 0.2 µm filter (Whatmann) with a 3% NaCl solution (w/v) and biocide ProClin 950 (Sigma-Aldrich) at a final concentration of 0.1% (v/v) to avoid bacterial growth, according to Fragoso et al. (2019). Yellow-green (505/515) fluorescence beads

(Molecular Probes, 1 μm) were used for size calibration before few consecutive runs. Fixed samples were transferred to a glass beaker and kept in suspension using a magnetic stirrer and injected into the CytoSense at a particle rate below 5000 particles s^{-1}. Each sample was run under two different protocols based on phytoplankton size: one to detect pico- and nanophytoplankton communities and another one for microphytoplankton, according to Hubert et al. (2024) . For picophytoplankton (in this study, roughly from 0.1 - 3 μm), the protocol was as following: sideward (SWS) trigger at 29mV, smart trigger of low detection threshold around 6 mV for total red fluorescence, medium photomultiplier tube (PMT) sensitivity, a low pump speed (5 μL s^{-1}) and a short sampling time (6 minutes). For nano- and microphytoplankton (> 3 μm), SWS trigger (29mV) with a higher detection threshold for total red fluorescence (16 mV), medium PMT sensitivity and a high pump speed (13 μL s^{-1}) and a long sampling time (8 minutes) was used.

Manual clustering (or 'gating' in flow cytometry studies) and identification of the main phytoplankton groups occurred based on light scattering and fluorescence properties of the particles, and in accordance to the interoperable vocabulary (Thyssen et al., 2022). Cytogram analyses were performed using CytoClus 4 software (Cytobuoy b.v., Netherlands). For simplicity, grouping of the phytoplankton functional groups were divided into size categories: pico- (~< 2 μM), nano- (~2-20 μM), and microphytoplankton (>20 μM).

3.7 Cyphonautes abundance from net samples

Identification and quantification of *M. membranacea* and *E. pilosa* cyphonautes from plankton net samples were performed at TBS using a Leica (model M205C) dissecting microscope. The EtOH-preserved samples were drained over a 110 μm mesh and rinsed with filtered seawater.

After rinsing, the samples were flushed into a plastic beaker for counting and diluted to a known volume. Subsamples of 3.5 mL were pipetted into a plexiglass counting chamber with four grooves, and the number of *M. membranacea* and *E. pilosa* larvae were counted. Subsamples were taken from each sample until 100 individuals of at least one of the species had been counted, or until the whole sample had been counted.

3.8 Cyphonautes size measurements from images

For cyphonautes size measurements, fixed zooplankton net samples were run using an imaging-recording microscope, called the AFTI-scope (Fig. 2a). The AFTI-scope is a flow through, open-source microscope designed to automatically capture images of particles in water samples inspired by the PlanktoScope design (Pollina et al., 2022). For more details about the AFTI-scope and how it was used to analyze cyphonautes larvae (see Hama, 2024 and Haugum et al., 2023).

A total of 21 net samples - triplicates of 7 sampling dates at station MI from February to June 2023 were run through the AFTI-scope after microscopic counts. Analyzing samples from February to early June gave the opportunity to compare the cyphonautes size during the kelp cultivation period.

Before running the preserved net samples in the AFTI-scope, samples were rinsed using a 110 µm mesh to remove the etOH and placed into a plastic beaker with filtered sea water with a known volume. All sample volumes in filter were run through the AFTI-scope. After the imaging record, images were analysed and manual annotation of length and width of larvae of *M. membranacea* or *E. pilosa* were performed using the image processing program ImageJ. Using the "line" tool in ImageJ, two lines were manually drawn in each larvae picture: one representing the antero-posterior basal axis (basal edge) and another one from oral to aboral axis (length) (Fig. 2b). An image of a microscopic ruler taken with the AFTI-scope in the same setup was used as size reference to set a scale of pixels per centimeter (pixels.cm^{-1}) for the corresponding frame. A total of 390 cyphonautes were measured.

3.9 Flatbed image scanning of kelp lamina

To get highly resolved images of the kelp lamina (Fig. 2c-h), a Canon flatbed scanner LiDE 400 with a scan resolution of 1200 DPI (= 47.24 Dots per mm) was used (Fig. 2d). An adhesive measurement tape was attached to the inside part of the scanner to generate a scale bar (Fig. 2e). Scans were made with the Canon IJ Scan Utility software, and the DPI was set to 1200 for high image resolution to detect the small bryozoan colonies, settlers and ancestrulae (minimum size diameter of 300 µm on the kelp lamina) (Fig. 2f-h). The files were stored as TIFF and later processed into SVG files.

For image acquisition, frozen specimens of *S. latissima* were thawed before scanning and cut into small parts so that the lamina pieces could fit into the scanner (Fig. 2c). More details about the scanning process is described in Snijder (2024).

3.10 Bryozoan image processing

A total of 1037 image scans, including both sides of the lamina, were made from 63 specimens. Bryozoan colonies, larval settlement and pre-ancestrulae on the images were manually marked with circles in the vector graphics editor program Inkscape (Inkscape Project, 2020). The pixel size of the kelp lamina and the marked bryozoan area (larvae, pre-ancestrula or colony) from the scans were calculated in the programming language Python™ (latest version used: 3.12.3 (Van Rossum and Drake Jr, 1995). The lamina area from the scans were retrieved by using the OpenCV library in Python (Bradski, 2000), where every scanned image was converted into grayscale and a threshold was made to create a binary mask, so the pixels of the lamina could be counted. Counts of the bryozoan colonies, settled larvae and pre-ancestrulae were also retrieved using Python, where every circle drawn in Inkscape was counted as one. For normalizing the data, the total counts were divided into the total scanned lamina surface area and calculated to counts per square meter area. For more details, see Snijder (2024).

3.11 Statistical analysis

All statistical analyses and graphs were conducted in Minitab 22.2.2 and Matlab R2024a using the *Statistics* and Machine Learning *Toolbox*. Cyphonautes larvae and phytoplankton abundances were checked for normality using a Shapiro-Wilk test and significant differences ($p < 0.05$) among years and stations were either tested using the two-sample t-test or Mann-Whitney, depending on normality. Boxplots indicate the median, the 25th and 75th percentiles, and outliers. To assess the environmental drivers of larval abundance, size, and colony counts of *E. pilosa* and *M. membranacea* in the water column and on kelp, we used generalized linear mixed-effects models (GLMMs) with a significance threshold of $p = 0.05$. Predictors included temperature, salinity, nutrient concentrations (PO_4^{3-}, NH_4^+, NO_3^-, $Si(OH)_4$), Chla_{CTD}, Turb$_{CTD}$, wind speed, above-water irradiance, and plankton size classes (pico-, nano-, microplankton). Seasonal variation was modeled using sine and cosine transforms of day-of-year. Larval size was analyzed separately for each species with environmental covariates and a random intercept for sampling day-of-year (YD). For larval abundance, random intercepts for station and year addressed spatial and temporal structure. Kelp-associated colony counts were modeled with a Poisson GLMM (log link), including environmental predictors and deployment as fixed effects, and specimen ID as a random intercept. Model predictions were compared to observed values using scatterplots and seasonal trends. Predicted and observed larval abundances were smoothed with LOWESS and visualized separately for 2022 and 2023.

4. Results

4.1 Time- series of environmental data

In both years, the average wind speed was high (many days above 10 m s^{-1}) in winter months, reducing to below 10 m s^{-1} from the end of March/beginning of April (Fig. 3a). This coincided with the continuous increase in photoperiod (increased daylight period from 100 – 200 µmol photons m^{-2} s^{-1} average daily light intensity (in PAR_{SAT} in March), sea surface temperature (SST > 6 °C) and the start of the spring phytoplankton bloom (values > 2 mg m^{-3}) (Fig. 3b-d). Storm events in spring, inferred by an increase in average wind speed again (> 10 m s^{-1}) and average reduction in average daylight due to clouds, occurred in the beginning of May in 2022 and in the end of May/beginning of June 2023 (Fig 3a,b).

Calm winds, in addition to increased daily PAR_{SAT} and increased sea surface temperatures (SST > 5°C) in April 2022 and 2023, appeared to be major meteorological factors that initiated the spring phytoplankton bloom (Fig 3a-d). Although the main phytoplankton peak (~ 5 – 6 mg m^{-3} Chla_{frame} concentration) was higher in April, a single peak was observed in 2023, while two bloom peaks were observed in 2022 - a smaller one in the end of March (~2 mg m^{-3}), followed by a larger one (~ 2 weeks) in mid-April (up to 6 mg m^{-3}) (Fig. 3d). From mid-May to mid-June, the Chla_{frame} concentration in the water appeared to be ~2 mg m^{-3} in both years, reflecting the high productivity of the area (Fig. 3d).

After the spring bloom peak in April, SST increases gradually until the end of June (from 6 °C to approximately 10- 12 °C), showing some peaks in warming periods, most related to the influence of weaker winds (Fig 3c). Surface water was generally colder in 2023 when compared to 2022 by 1-2 °C.

Values for Turb$_{frame}$ varied between years, being higher in 2022 and reflecting the peak of the spring bloom in 2022 (0.35 FTU), while it peaked in March 2023 (> 0.4 FTU) and increased from April to July (from 0.1 to 0.2 FTU), reflecting not only phytoplankton but other particles in the waters (marine snow and sediments) (Fig. 3e).

4.2 Environmental data from vertical profiles

Counter plots of temperature and salinity from vertical profiles of the CTD sensor showed similar trends in temperature distribution within the upper 10-15 m in both stations and years (Fig. 4a-d). Stratification increased from early May onwards, with a rapid increase in temperature in the upper 10 m from May (~ 8 °C) towards the end of June (> 10 °C) (Fig. 4a-d). Salinity, however, varied between sites and with time, being generally fresher at station MI than at MO and in 2022 than in 2023, particularly in late March and mid-May (salinity down to 32) (Fig. 4e-h).

4.3 Nutrient concentrations

Nutrient concentrations did not vary between years and sites ($p < 0.05$, t-test), except for ammonium ($p = 0.0001$, Mann-Whitney test), which was higher in 2022 ($\mu = 0.8$ µM) than in 2023 ($\mu = 0.2$ µM). Nutrient concentrations followed a general trend typically observed for temperate Northeastern Atlantic waters (except for NH_4^+ in 2023), being high during February, and decreasing with time (particularly during the spring bloom in the end of April) (Fig. 5). After the first bloom peak in April, nutrients bounced slightly back up in May 2022 and June 2023 (Fig. 5), possibly because of strong winds and mixing (see Fig. 3a). In general, NO_3^- concentrations were higher in station MO (up to 4 µM higher in some cases) compared to MI and in 2022 (up to 10µM) and compared to 2023 (up to 5µM) (Fig. 5a,b). For PO_4^{3-} concentrations, a similar trend was observed, being higher at stations MO (up to 0.1 µM higher) than station MI (Fig. 5c,d). During winter 2022 (February and March), NH_4^+ concentrations were exceptionally high (> 1 µM) and possibly influenced by the first bloom peak in March in 2022 (Fig. 5e,f). $SiOH_4$ concentrations followed similar patterns when compared to PO_4^{3-} and NO_3^-, being high before the bloom (~ 3µM), at a storm event in May 2022 and June 2023 (~2 µM) and station MO (up to 0.1 µM higher) than station MI (Fig. 5g,h).

4.4 Phytoplankton size structure

Phytoplankton size-structure varied in time and between stations in 2022 and 2023 (Fig. 6). Abundances of pico-, nano- and microphytoplankton were not significantly different between years and sites (p > 0.05, two-samples T-test). In general phytoplankton abundances were as expected low in winter months (< 40 ind.µL^{-1} in February and early March), increasing with time and peaking on 3rd June 2022 at stations MI and MO and 23rd May 2023 (abundances > 100 ind.µL^{-1}) (Fig. 6). Picophytoplankton and nanoplankton were more abundant in late May and June 2022 and mid-May 2023. Picoplankton contributed to the largest fraction of the phytoplankton, reaching > 65% on 3rd June 2022 (for both MI and MO stations), followed by nanophytoplankton > 58% on 20th April 2022 and 18th April 2023 (particularly at station MI) (Fig. 6). The microphytoplankton population contributed to an average of 13% of phytoplankton for both locations and years, peaking on 22nd March 2022 and 18th April 2023 (Fig. 6).

4.5 Cyphonautes abundance and size distributions

Abundance of cyphonautes larvae from stations MI and MO was highly variable among replicates (Fig. 7a,b), although not significantly different among years for *M. membranacea* and *E. pilosa* (p= 0.635 and p= 0.184, respectively, two-samples T-test) and stations (p= 0.725 and p= 0.932, respectively, two-samples T-test). However, a general trend was observed with *M. membranacea* larval stages in higher concentrations in most cases, particularly in 2022 (Fig. 7a). Two peaks in abundance of cyphonautes were observed for *M. membranacea* in both years: early April (up to 177 ind m^{-3} in station MI) and early June in 2022 (values up to 341 ind m^{-3} at station MO), while in 2023, the peak was observed in mid- to late April (values for *E. pilosa* cyphonautes up to 403 ind m^{-3}) and late June in 2023 (values up to 444 ind m^{-3} for *M. membranacea*) (Fig. 7a,b).

Cyphonautes were also highly variable in size, ranging from 0.065 – 0.452 mm and 0.07 – 0.417 mm in maximum basal edge and length, respectively for *E. pilosa* larval stages (n=165) (Fig. 7c,d). For *M. membranacea* larval stages, the sizes ranged from 0.06 – 0.752 mm in maximum basal edge and 0.075 – 0.605 mm for length (n=225) (Fig. 7c,d). There was a general increase in size from late May until late June, particularly for *M. membranacea* cyphonautes (Fig. 7c,d). The median value of basal edge and length were ~0.15 mm in mid-March (n= 22), while in early June, these values were 0.65 mm and 0.52mm (n=30), respectively (Fig. 7c,d). In late June, however, the values of median basal edge and length (n=10) were reduced to 0.43 mm and 0.35 mm, respectively (Fig. 7c,d).

4.6 Bryozoan recruitment and colony coverage on the kelp lamina

Replicate samples of *S. latissima* sporophytes were collected across seven sampling days (25^{th} April, 10^{th} May, 15^{th} May, 23^{th} May, 7^{th} Jun, 15^{th} Jun, 29^{th} Jun) to register bryozoan coverage on the lamina. Sporophyte samples were taken randomly from the cultivation line for two different deployment dates (October 2022 and January 2023) at station MI. *Saccharina latissima* from both deployment lines had small percentage bryozoan coverage (10^{-5} to 10^{-2} % cover) from late April to early May (Fig. 8a). However, the percentage coverage increased rapidly in the middle of June, with a higher percentage of coverage for the October deployment line (average of 11% coverage) than the January deployment (average of 3% coverage) on 29^{th} June 2023 (Fig. 8a). The first settlement was detected on 25^{th} April 2023, with a total of 6 settled larvae from samples taken that day. No clear colonies were detected at that stage (Fig. 8b,c). Two peaks in the total settlement and pre-ancestrulae counts occurred for both lines on the 15 May 2023 and the 15 June 2023 (187 and 826 counts per m^2 for October, respectively and 126 and 618 per m^2 for January) (Fig. 8b). Almost all settlers and pre-ancestrulae were found on the meristem, the newer part of the lamina, for the whole sampling period (Snijder, personal observation). The total counts of bryozoan colonies m^{-2} in *S. latissima* increased gradually and showed the same pattern of variation for both deployment lines, with the highest number of colonies on 29th June 2023 (approximately 1260 counts per m^2 for October and 860 for January, Fig. 8c). For June, bigger colonies (e.g. > 5cm) were mainly found at the older part (tip) of the lamina (data not shown).

In total, 1.3307×10^5 colonies of settled larvae and pre-ancestrulae were detected on the scanned images, from which 3803 counts were from the replicates from the line deployed in January 2023 and 9.504×10^4 counts from the line deployed in October 2022. Bryozoan colony size increased in time, showing a logarithmic trend of increase (Fig. 8d). The largest colonies (between 10^3 to 10^4 mm^2) were only found on the 29th June 2023 with the lowest abundance (Fig. 8d). The highest abundance of colony sizes was found in the smallest size range (0 - 0.5 log mm^2) for this date as well (Fig. 8d).

4.7 General linear mixed models (GLMM)

GLMM was fitted to log-transformed larval abundance of *E. pilosa* and *M. membranacea* using environmental and biological predictors, with random intercepts for Station and Year. Several predictors were statistically significant ($p < 0.05$) (Table 1). Seasonal patterns captured by the sine ($\sin(2\pi\ YD/365)$ or SinYD) and cosine ($\cos(2\pi\ YD/365)$ or CosYD) of the day of year (SinYD: $p = 0.0003$; CosYD: $p = 0.0008$) were strong drivers of abundance. Nutrient concentrations, particularly phosphate (PO_4^{3-}; $p = 0.0002$) and NH_4^+ ($p = 0.002$), and light (PAR_{SAT}; $p = 0.003$), pico-phytoplankton abundance ($p = 0.01$) and salinity ($p = 0.0002$) had significant negative effects (Table 1).

For *M. membranacea* larval abundance, the GLMM showed that significant seasonal variation was evident, with both the sine and cosine components of the year day (YD) (SinYD: $p = 0.00002$; CosYD: $p = 0.007$) showing strong effects (Table 1). Among environmental predictors, salinity was also a strong negative predictor ($p = 0.0001$), NH_4^+ ($p = 0.04$), and PAR_{SAT} ($p = 0.04$) also had a significant negative association with larval abundance of *M. membranacea*.

From all environmental and biological factors, none were statistically significant predictors *E. pilosa* larvae size ($p > 0.05$). For *M. membranacea* larval size, salinity ($p = 0.01$), Chla_{frame} ($p = 0.0004$) and wind strength ($p = 0.005$) had significant negative effects on larval size, while $Turb_{frame}$ ($p = 0.00002$) and PAR_{SAT} ($p = 0.00008$) had positive effects (Table 1).

For bryozoan settlement and colony counts, the results of the model show that deployment timing ($p < 0.0001$), temperature ($p < 0.0001$), and Chla_{frame} ($p = 0.01$) were strong predictors (Table 1).

5. Discussion

5.1 Phytoplankton and cyphonaute larvae dynamics

The onset of spring phytoplankton blooms in the Frohavet region are affected by weather, particularly calm winds and cloud-free days (Fragoso et al., 2024). In this study, the phytoplankton bloom started earlier in 2022 (late March) than in 2023 (early April), peaked in mid-April for both years and persisted with high Chla_{frame} concentrations (~2 mg m^{-3}) throughout the end of June (Fig. 3c). Likewise, cyphonautes showed two peaks in abundance- one in April/May and a second one in June/July, with the first peak occurring 1–2 weeks after the phytoplankton bloom onset. Previous studies reported varying results, with high abundance of *M. membranacea* larvae from January to July in southern California (Yoshioka, 1982), while peak cyphonautes abundance occurred later in other regions: June in mid-Norway and in September in Nova Scotia (Førde et al., 2016; Saunders and Metaxas, 2010).

High abundance of cyphonautes have usually been found in shallow, warm and low-salinity surface layers, when a strong pycnocline was present, indicating potential onshore transport of the larvae during wind-driven downwelling events (Saunders and Metaxas, 2010). In this study, relatively low salinity (~ 32) was a strong predictor for high abundance of cyphonautes in the water, when thermal and haline-stratification was strongest in June 2022 and July 2023. However, other studies showed the presence of the cyphonautes was most abundant in the upper 5-10 m of the water column during the cooler winter months and most scarce in high temperature, surface waters during summer in California (Yoshioka, 1982). Variability in sampling timing, temperature and frequency could explain the discrepancy in these studies, as well as small-scale patchiness, typical in zooplankton communities in coastal regions, including kelp beds (Fragoso et al., 2022; Maldonado-Aguilar et al., 2023).

While most bryozoan larvae are lecithotrophic (non-feeding and yolk-dependent), cyphonautes larvae of *M. membranacea* and *E. pilosa* are obligatory planktotrophs (feed on plankton) with a functional digestive tract (Woollacott and Zimmer, 2013). This means that these larvae rely on feeding on plankton and other particles and continue to develop in the water, where they can have a long pelagic phase, lasting days to weeks, and potentially as long as two months (Woollacott and Zimmer, 2013). Similarly to other studies found in this region (Fragoso et al., 2024, 2021), Chla_{frame} concentration was relatively high (> 2 mg m^{-3}) from April until late June and could reflect a source of food that can keep cyphonautes larvae in the water for longer periods. Moreover, in this study, the abundance of *M. membranacae* and *E. pilosa* larvae had a negative association with Chla_{frame} concentrations and picoplankton abundance, respectively, possibly indicating of clearance rates of picoplankton by the larvae as well as other zooplankton. While starvation can reduce internal structures of the organism needed for competence and metamorphosis (pyriform organ, internal sac), their shell size and ciliary bands for swimming and feeding are unaffected, allowing them to maintain mobility and survive (Strathmann et al., 2008). Enhanced stratification and northward flow of the NCC, typically increased from spring to late summer, can potentially have a role in the widespread dispersal of the larvae over large areas, allowing them to colonize more diverse habitats and expand their geographical area before reaching metamorphic competence.

5.2 Environmental effects on larval competence, settlement and colony growth

To be able to settle on a kelp lamina, cyphonautes need to reach a certain size for competence before metamorphosis occurs (750 µm in length for *M. membranacea* and 400 µm for *E. pilosa* , Ryland and Stebbing 1971). In this study, *E. pilosa* larvae from plankton nets were relatively small (0.24 mm basal edge, 0.27 mm height) compared to values from the literature (0.39 mm across the base and 0.36 mm in height) (Atkins, 1955), and are presumably in the pre-competent stage (< 440 µm, according to Saunders and Metaxas, (2010). *M. membranacea* larvae were also smaller (basal edge of 0.62 mm) compared to that found in the literature (0.75 - 0.8 mm basal edge), but size generally increased by late May/early June, reflecting later stages of development before settling (Atkins, 1955; Ryland and Stebbing, 1971). The variability in size of *M. membranacea* larvae found in our study suggested that a second spawning event possibly occurred in June, likely from colonies already established at the kelp farm or from surrounding wild kelp. Fast competence and presence of multiple cohorts of *M. membranacea* larvae have been suggested as successful traits for settlement in ephemeral habitats, contributing to the invasive potential of this species *along* the east coast of the US (Saunders and Metaxas, 2010).

Temperature variability is known to affect the growth and development rates of larvae of many marine animals, directly influencing their dispersal distance and survival (O'Connor et al., 2007). In this study, high temperature, low salinity and low Chla_{frame} concentrations were found to be predictors of *M. membranacea* size, while for *E. pilosa*, no significant effect of the environmental conditions on larval size was found. The fact that cyphonautes were found in high abundance in the water but did not settle on the seaweed earlier in the season (early April) suggested that the temperature was not high enough for the larvae to reach competence and, thus, to settle. In this study, the first bryozoan settlement was detected on 25$^{\mathrm{th}}$ April at 6.6 °C, with peaks in larval settlement and pre-ancestrulae counts following sharp temperature increases: from 7 °C to ~9 °C, and later from 9.5 °C to 11.5 °C. In other regions, the timing of first bryozoan colony appearance has been reported to occur when temperatures were 7.5 °C in early May (Njåstad, 2018) and 10 °C in June in Central Norway (Førde et al., 2016), while colonies appeared at 6–8 °C in mid-July in Tromsø (Matsson et al., 2019). The influence of temperature on cyphonautes settling, recruitment and colony growth has been consistently reported in the literature in subarctic regions (Caines and Gagnon, 2012; Saunders and Metaxas, 2007, 2010, 2009; Scheibling and Gagnon, 2009), with models predicting that a 1 °C or 2 °C rise could raise kelp coverage by a factor of 9 and 62, respectively (Saunders et al., 2010). In St. Margarets Bay, Nova Scotia, for example, *M. membranacea* settler abundance was strongly associated with growing degree-days (GDD)—a measure of thermal history—and was 10 × higher following a significantly warmer winter (Saunders and Metaxas, 2007).

High exposure, likewise, could influence recruitment, because strong currents could hinder cyphonautes settlement on the kelp. Intermediate wave exposure in Nova Scotia has been suggested to correlate positively with high numbers of *M. membranacea* settlers on kelp lamina (Caines and Gagnon, 2012). In this study, a storm taking place in early June 2023 (high wind speed, up to 20 m s^{-1}) was coincident with intense water mixing, low larval abundances in net samples and low larval settlement and pre-ancestrulae abundance in kelp blades. Visch et al., (2020) found that kelp biofouling declined with increasing wave exposure, dropping from 10% at sheltered sites and 6% at moderately exposed sites to just 3% at exposed locations. This contrasts with Matsson et al. (2019), who observed coverage by *M. membranacea* colonies 64 × higher in more exposed, semi-offshore site when compared to an inshore, fjord site. For both studies, the variation in coverage was high between the locations, even if the distance between locations was small: 20 km away for Visch et al. (2020) and 50 km away for Matsson et al. (2019).

Membranipora membranacea larvae use their pyriform organ and mucus to crawl along the kelp lamina, typically moving from older to younger tissue, more proximal regions for settlement (Matson et al., 2010). While the mechanisms guiding this site selection remain unclear, our study found that nearly all larvae and pre-ancestrulae settled on the newer, proximal lamina, indicating a true preference rather than gregariousness or space limitation. Similar patterns have been reported previously (Denley et al., 2014; Matsson et al., 2019), although *E. pilosa* settlers were more abundant on slightly older tissue, likely due to competition with *M. membranacea* (Denley et al., 2014). Heavier fouling was observed on older lamina regions, possibly driving larvae to settle on newer tissue where space is available, and growth time is maximized before tip shedding (Matsson et al., 2019).

The percentage coverage was higher from the October 2022 deployment than from the February 2023 one. Also, the colony abundance was higher for October with a greater number of larger colonies. Larger colonies, which grew exponentially and thus expand in area more rapidly than smaller ones, had higher coverage (Hermansen et al., 2001; Riisgård and Goldson, 1997; Saunders and Metaxas, 2007). Moreover, the October sporophytes were longer, providing more substrate area for the colonies to expand. Matsson et al. (2021) also observed that earlier deployment times (February compared with April and May) gave a higher biomass of *S. latissima*, and higher biofouling coverage.

Bryozoan colonies need an ambient water flow to receive optimal food supply and intake for growth. For example, water velocity is typically higher in shallower water than in deeper water, and increased water velocity in near-surface kelp farms can enhance food supply and growth rates of filter-feeders (Lenihan et al., 1996). Wilson-Ormond et al. (1997) found that low water velocities limited the flow of food particle flux and slowed oyster growth, suggesting that water flow was more influential than food concentrations. On the other hand, Hermansen et al. (2001) observed no difference in colony growth when *E. pilosa* was exposed to increased water flow velocities (observations from 30 cm s^{-1} up to 150 cm s^{-1}). Saunders and Metaxas (2009) reported an interaction between temperature and food availability, where food limitation reduced colony diameter only at higher temperatures, likely due to elevated metabolic demands. This indicates the existence of inter-dependent relationships between environmental variables (temperature, Chl *a*, current velocity, light) and bryozoan feeding, which needs to be better understood.

6. Conclusions

This study highlights the complex interplay of environmental and biological variables driving bryozoan biofouling at a kelp farm located in Central Norway. While *M. membranacea* and *E. pilosa* larvae peaked in abundance during spring, larval size for *M. membranacea* in the water —rather than abundance alone—was more closely linked to colony settlement, emphasizing the role of larval maturity in biofouling dynamics. Larval abundance was high in stratified, low-salinity waters. Recruitment was greater on kelp deployed earlier in the season, with substantial increases in colony size and coverage observed from May to June. Key drivers of biofouling were food availability following spring phytoplankton blooms, stable stratified waters, and sudden temperature rises. Such factors influenced larval abundance, settlement timing, and colony development of *M. membranacea* and *E. pilosa* on the kelp.

7. Declaration of competing interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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10. Declaration of generative AI and AI-assisted technologies in the writing process.

During the preparation of this work the G.M.F. used ChatGPT, in certain cases, to improve the quality of the text and assist with coding writing in Matlab. After using this tool, G.M.F. reviewed and edited the content as needed and takes full responsibility for the content of the published article.

11. Author contributions: CRediT

Conceptualization: G.M.F., D.A. Data curation: N.H., E.S., M.H. Formal analysis: G.M.F., N.H., E.S., Funding acquisition: G.M.F., D.A., G.J., Supervision: G.M.F., D.A., A.R.B-S., O.J., Y.O., G.J., Writing – original draft: G.M.F., N.H., E.S., D.A., O.J., Writing – review and editing: all authors.

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Table 1. Results from the General Linear Mixed Models. ****p <0.001; *p <0.05**

	E. pilosa larvae abundance. AIC: 222.23, BIC: 266.77, LogLikelihood: -92.117, Deviance: 184.23.				*M. membranacea* larvae abundance. AIC: 236.01, BIC: 280.54. LogLikelihood: -99.006, Deviance: 198.01.			
Term	**Est.**	**SE**	**DF**	**pValue**	**Est.**	**SE**	**DF**	**pValue**
Intercept	72.33	19.00	61	****0.0003**	72.59	20.77	61	****0.0009**
SinYD	5.10	1.32	61	****0.0003**	6.67	1.44	61	****0.00002**
CosYD	-2.34	0.67	61	****0.0008**	-2.04	0.73	61	****0.007**
Pico	-0.03	0.01	61	***0.01**	0.01	0.01	61	0.3
Nano	0.03	0.02	61	0.2	0.04	0.03	61	0.1
Micro	0.01	0.08	61	0.9	-0.02	0.09	61	0.8
PO_4	-6.95	1.78	61	****0.0002**	-3.19	1.94	61	0.1
NH_4	-1.32	0.41	61	***0.002**	-0.92	0.45	61	***0.04**
NO_3	0.18	0.15	61	0.2	0.28	0.16	61	0.1
$SiOH_4$	-0.17	0.29	61	0.5	-0.62	0.31	61	0.1
Temperature	0.10	0.26	61	0.7	0.67	0.29	61	0.02
Salinity	-2.06	0.52	61	****0.0002**	-2.29	0.56	61	****0.0001**
Chla_{CTD}	-0.10	0.13	61	0.5	-0.20	0.14	61	0.2
$Turb_{CTD}$	-5.12	2.54	61	***0.05**	-1.51	2.78	61	0.6
Wind	0.02	0.09	61	0.8	0.08	0.10	61	0.4
PAR_{SAT}	-0.01	0.002	61	***0.003**	-0.01	0.002	61	***0.04**

	E. pilosa larvae size AIC: 222.23, BIC: 266.77 LogLikelihood: -92.117, Deviance: 184.23.				*M. membranacea* larvae size AIC: -457.55 , BIC: -426.8 LogLikelihood: 237.77, Deviance: -475.55			
Term	**Estimate**	**SE**	**DF**	***p*Value**	**Estimate**	**SE**	**DF**	***p*Value**
Intercept	-13.93	26.33	158	0.6	52.00	20.92	218	***0.01**
Temperature	-0.06	0.11	158	0.6	0.23	0.09	218	***0.007**
Salinity	0.41	0.80	158	0.6	-1.58	0.63	218	***0.01**
Chla_{frame}	-0.04	0.32	158	0.9	-0.97	0.27	218	****0.0004**
$Turb_{frame}$	2.44	1.56	158	0.1	7.34	1.67	218	****0.00002**
Wind	0.05	0.05	158	0.3	-0.13	0.05	218	***0.005**
PAR_{SAT}	0.001	0.002	158	0.7	0.01	0.002	218	****0.00008**

Term	**Estimate**	**SE**	**DF**	***p*Value**
Intercept	-5.05	1.27	58	****0.0002**
Deployment	0.20	0.01	58	****<0.0001**
Temperature	0.93	0.10	58	****<0.0001**
$Turb_{frame}$	-0.40	4.41	58	0.9
Chla_{frame}	1.14	0.47	58	***0.01**

Figure 1

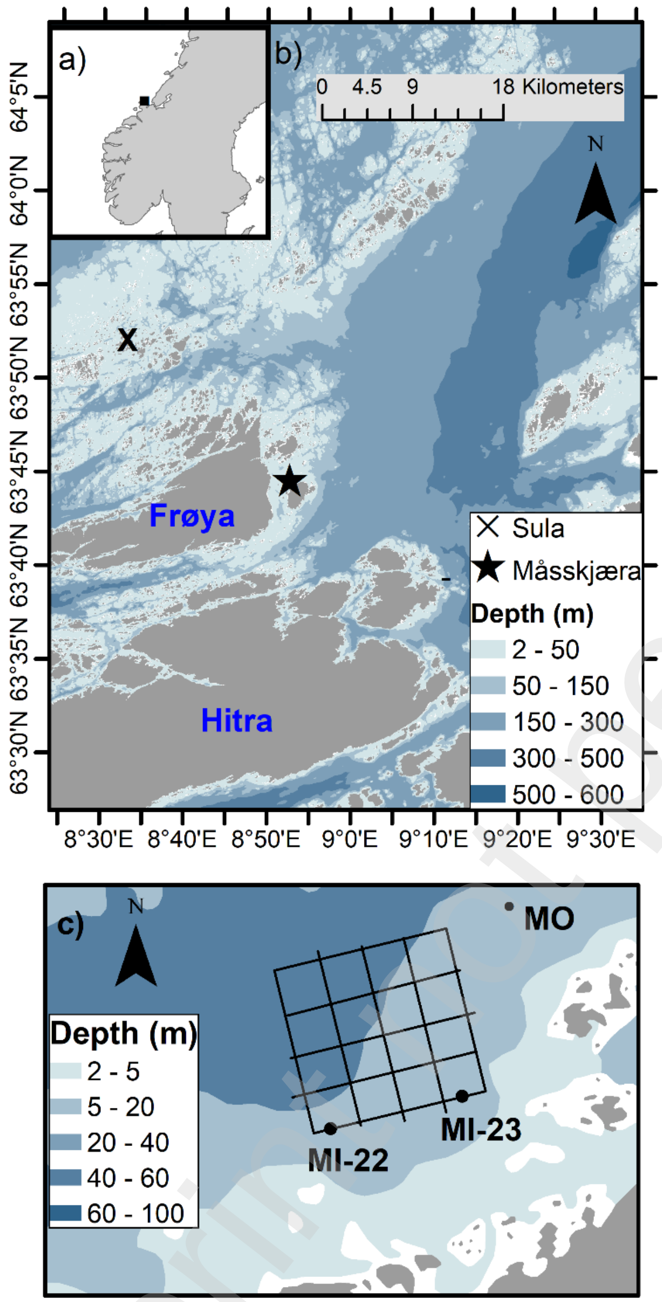


Figure 1: a) Map of the geographical location of Måsskjæra kelp farm located off the coast of central Norway. b) The location of the seaweed farm, including Sula meteorological station (star symbol), where wind data were collected. c) Illustrations of the layout of the seaweed farm and the sampling sites for environmental data and cyphonautes larvae abundance (Stations MI in 2022 and 2023 and MO – same station for 2022/3).

Figure 2

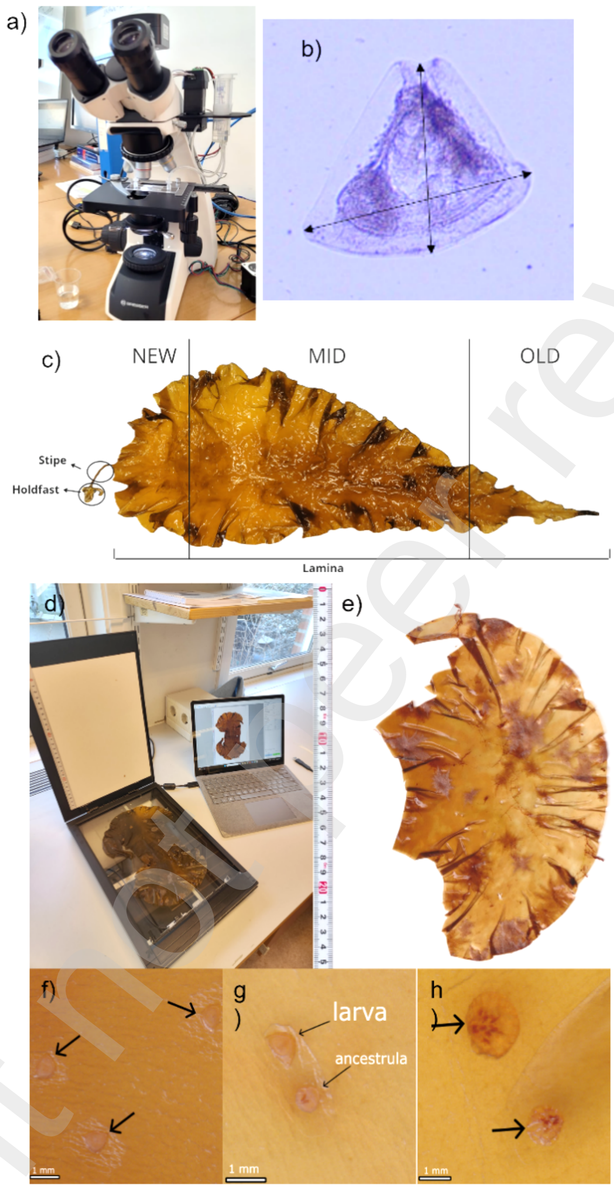


Figure 2: a) The AFTI-scope scope assembled on the Bresser Trino TFM-301 microscope. b) Lines showing the size measurements (basal edge and length) extracted from cyphonautes larvae individuals – antero-posterior basal axis (basel edge, horizontal line) and oral-aboral axis (length, vertical line). c) An image of a sporophyte with lamina, stipe and holdfast and an example showing the divisions of the kelp lamina into new (meristem), middle and old tissue (tip). d) The Canon A4 paper-size flatbed scanner LiDE 400 with kelp and (e) one example of an image with a scale bar after scanning. f) Examples of three settled larvae (note the triangular shell shape), g) a settled larva and a pre-ancestrula stage, where the septum is clearly observed and h) a developed small colony (approximate spatial resolution of 21 µm).

Figure 3

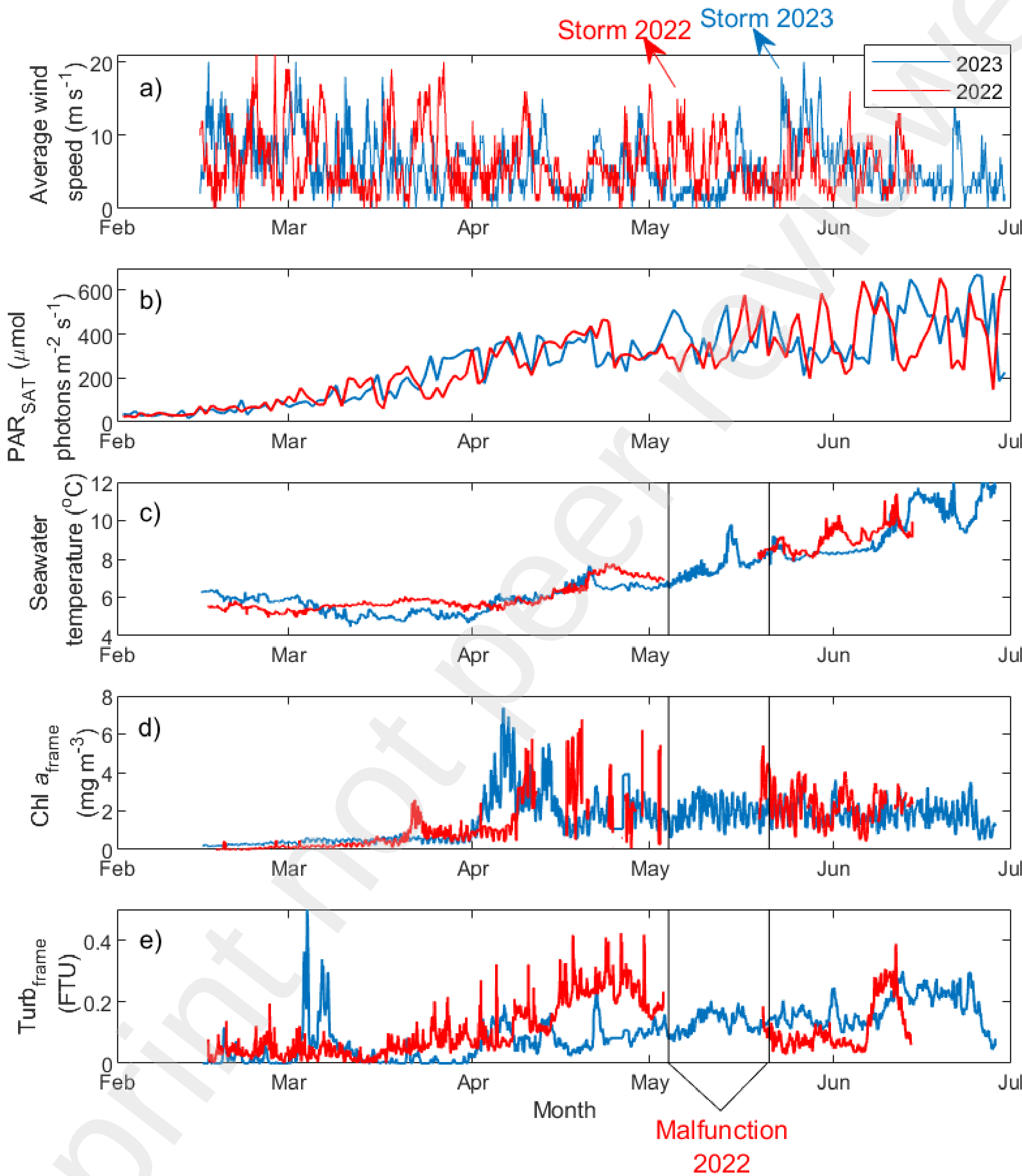


Figure 3: a) Time series data of a) average wind speed (m s^{-1}), b) average daily irradiance (PAR$_{SAT}$, µmol photons m^{-2} s^{-1}), c) seawater temperature (°C), d) chlorophyll a fluorescence (Chla_{frame} mg m^{-3}) and e) turbidity (Turb$_{frame}$, FTU) collected every 10-min using the C3 submersible fluorometer at Måsskjæra farm (station MI, inside the farm) from mid-February to mid (2022, red) and end of June (2023, blue).

Figure 4

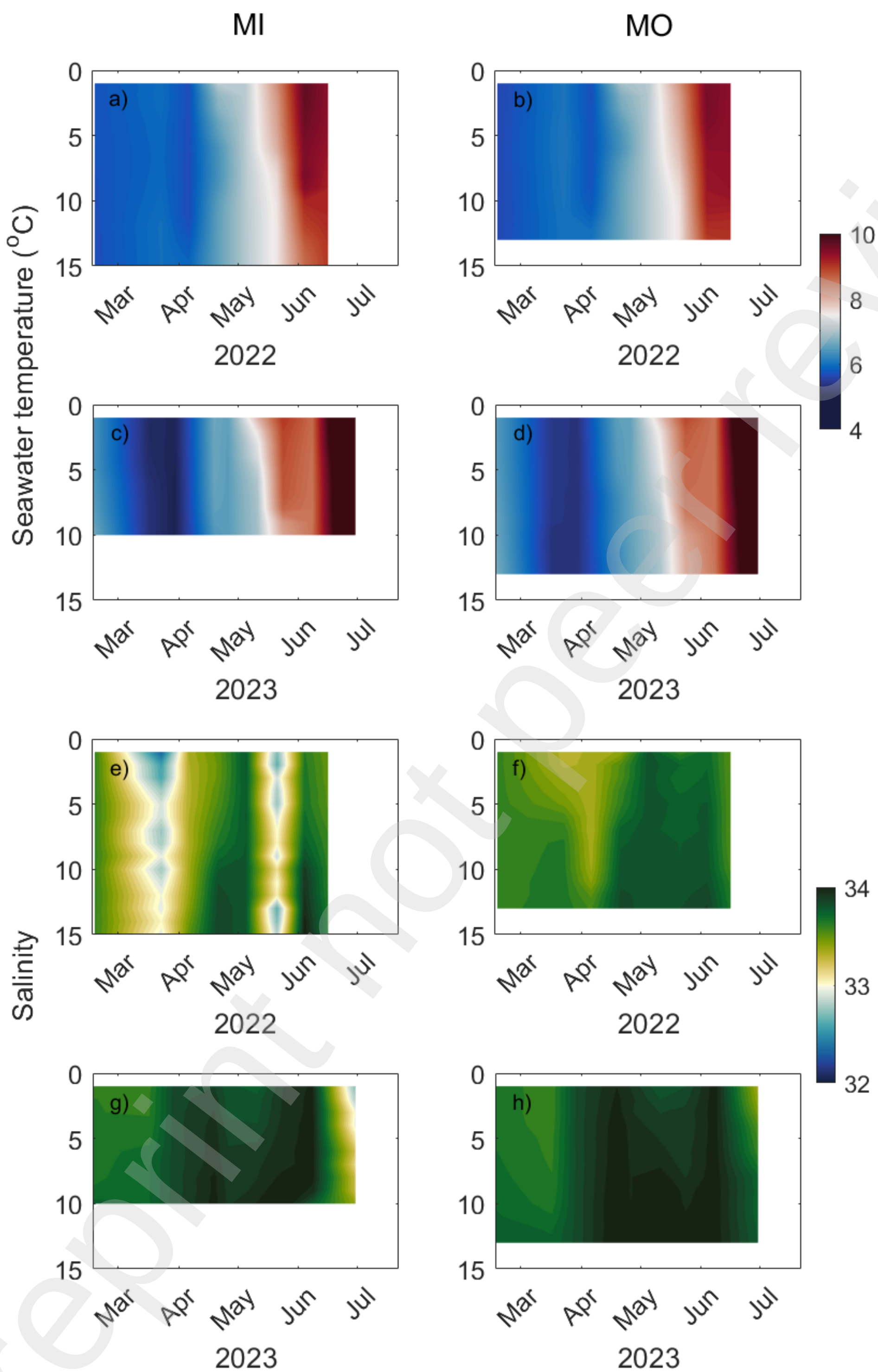


Figure 4. Counterplots of (a-d) temperature and (e-h) salinity from vertical CTD profiles at stations MI (left) and MO (right) during 2022 and 2023.

Figure 5

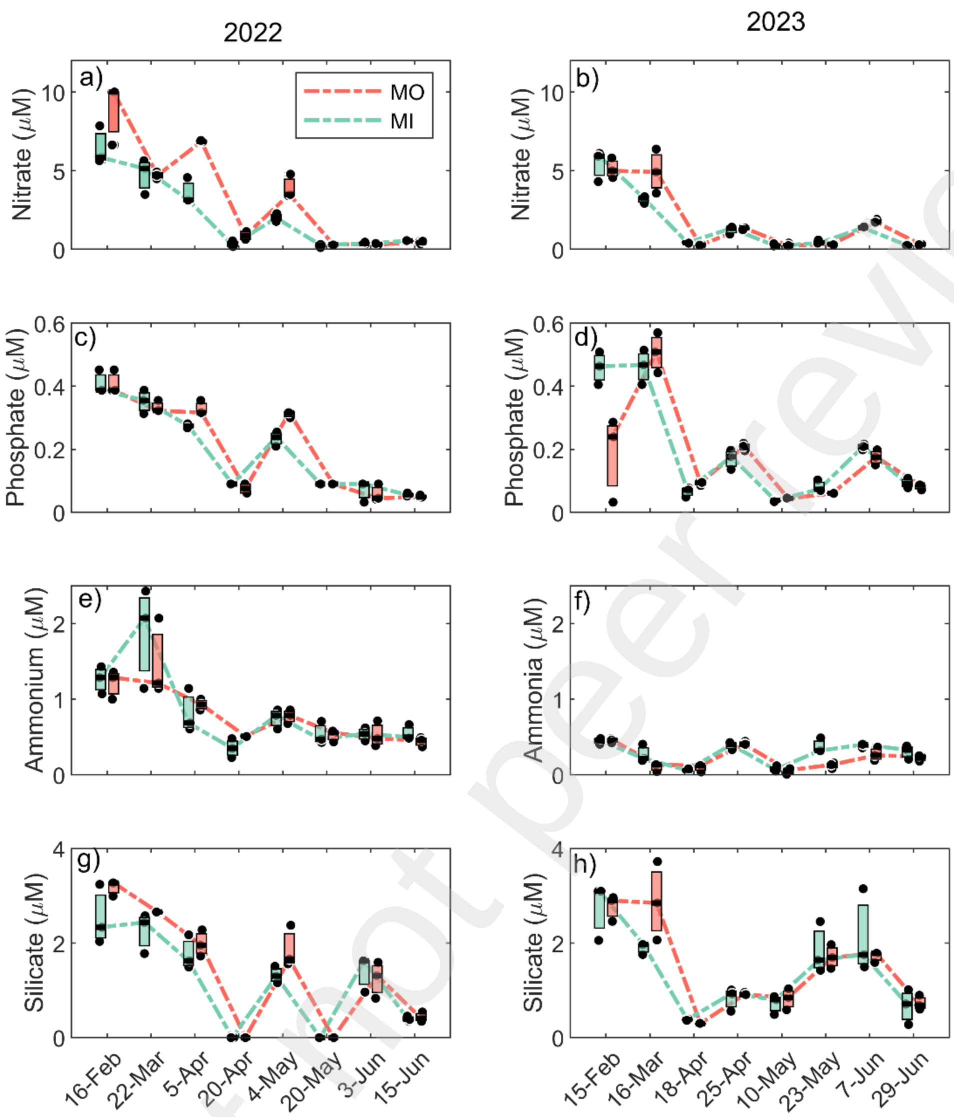
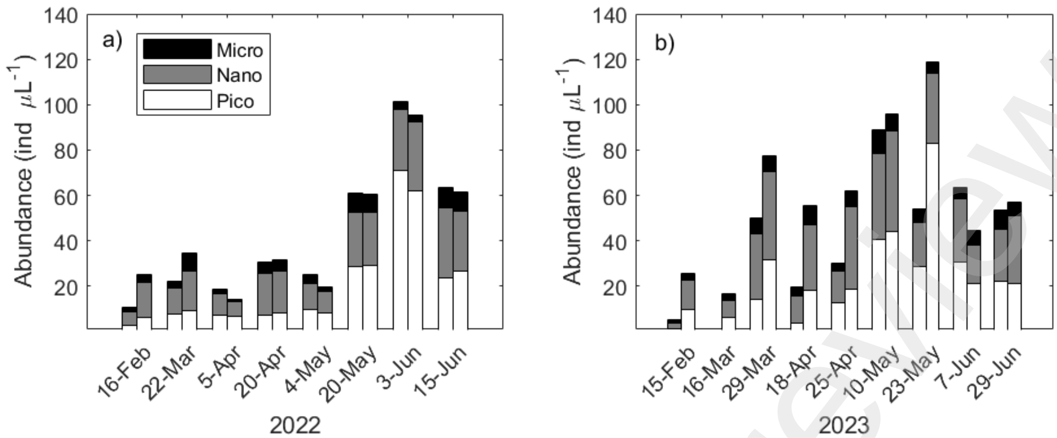


Figure 5. Nutrient concentrations (nitrate, phosphate, ammonium and silicate in µM) collected every 2 weeks from February to June in 2022 (left) and 2023 (right) from MI (green) and MO (orange) stations. Dots represent replicates.

Figure 6



Figue 6. Time series of phytoplankton abundances (ind μL^{-1}) based on different size: picoplankton (white), nanoplankton (gray) and microplankton (black) for a) 2022 and b) 2023. For each day, the first bar represents MI stations, while second bar represents MO stations.

Figure 7

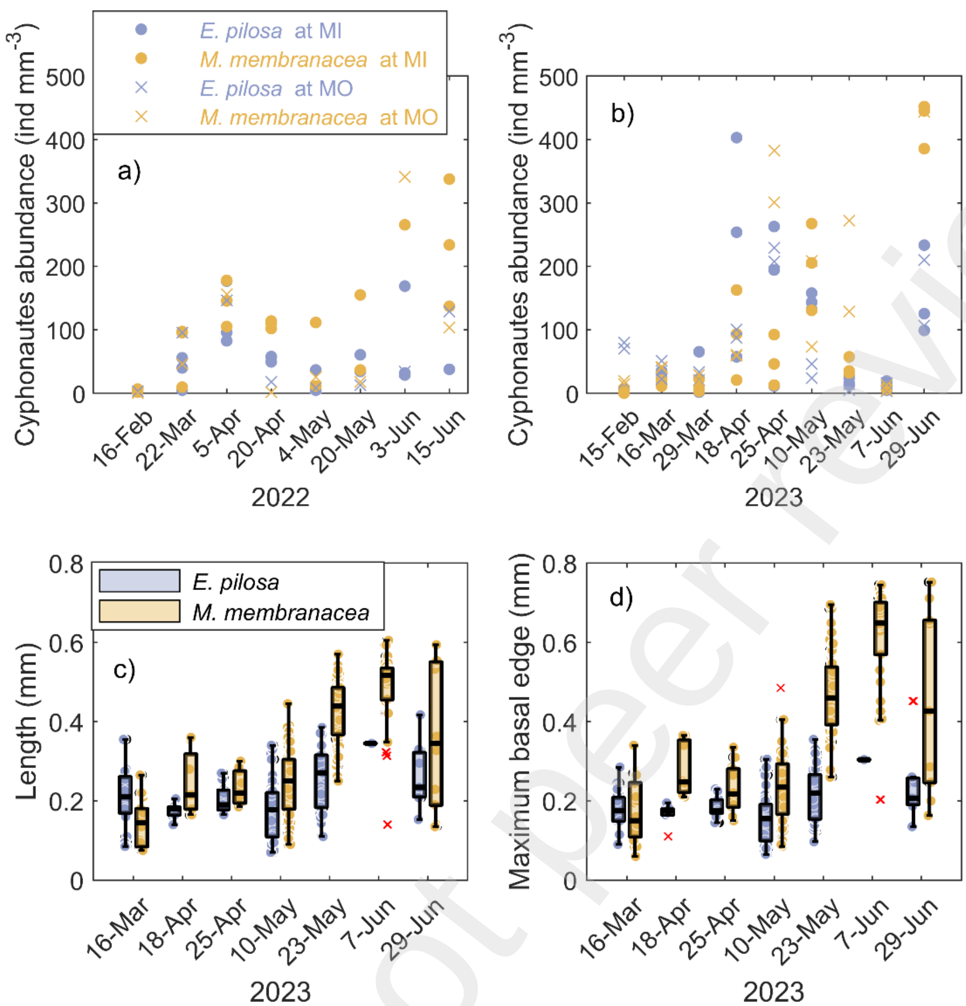


Figure 7. Larval abundance (cyphonautes) of *Electra pilosa* (purple) and *Membranipora menbranacea* (orange) at stations MI (filled circle symbol) and MO (cross symbol) collected every 2 weeks from February to June in a) 2022 and b) 2023. Box plot of larval size of *E. pilosa* (purple) and *M. membranacea* (orange), including c) oral-aboral axis (length, µm) and d) antero-posterior basal axis (maximum basal edge, µm) measured in 2023 at station MI only.

Figure 8

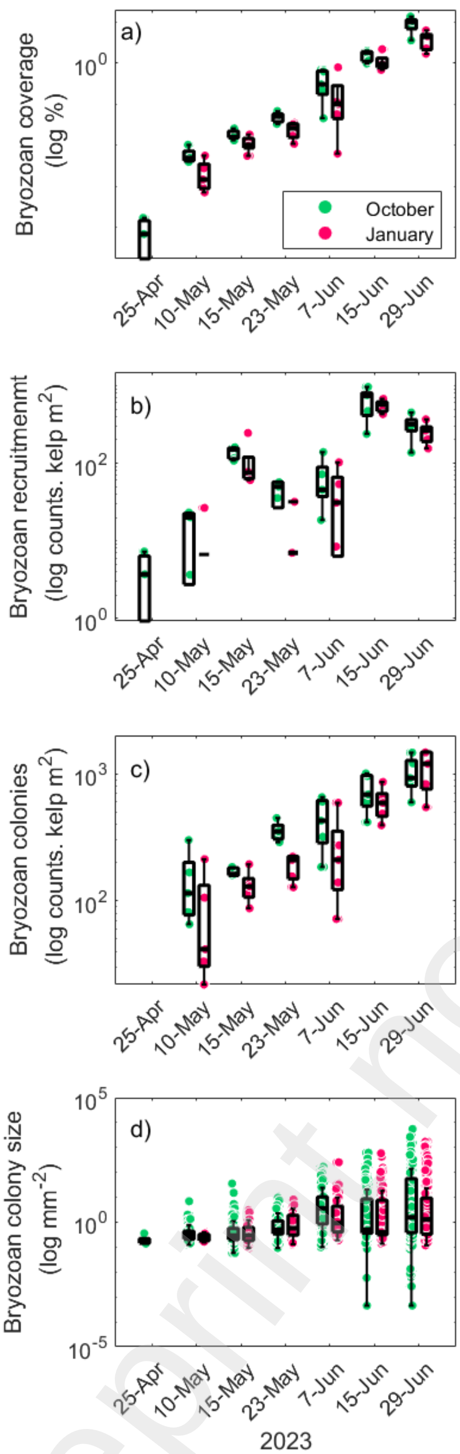


Figure 8 – Time series of bryozoan a) percentage coverage (%), b) larval settlement and pre-ancestrulae stage per kelp lamina, m^{-2}, c) colony counts per kelp lamina, m^{-2}, and d) colony size (m^2) in logarithmic scale for samples harvested in 2023 and deployed in October 2022 (green) and in January 2023 (pink).

Figure 9

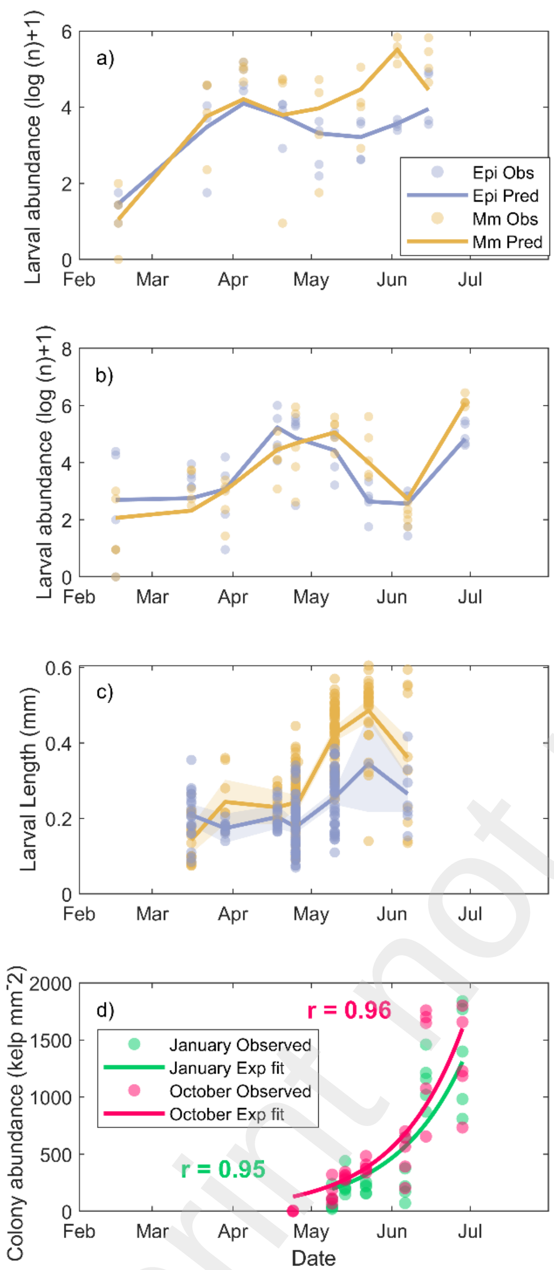


Figure 9. Observed and predicted seasonal trends in larval abundances of *Electra pilosa* (blue) and *Membranipora membranacea* (orange) from a) 2022 and b) 2023 as well as c) larval size (length) from the water column as well as d) colonization in the kelp in 2023 using generalized linear mixed-effects models (GLMMs). Predictions were generated using full models incorporating seasonal (sine/cosine of day of year), biological, and physicochemical covariates, with random intercepts for station and year (see method section).

Table S1. Sampling dates (in 2023) for CTD measurements (10 m) and water sampling for dissolved inorganic nutrients. Sampling dates for collection of *S. latissima* replicates for both the deployment lines (October 2022 and January 2023) are given.

Date	CTD and water sampling	Replicate nr. from the line deployed in Oct '22	Replicate nr. from the line deployed in Jan '23
15-02	X	-	-
16-03	X	-	-
29-03	X	-	-
18-04	X	-	-
25-04	X	3	-
10-05	X	5	5
15-05	-	5	5
23-05	X	5	5
07-06	X	5	5
15-06	-	5	5
29-06	X	5	5